ThreadSync – Project BRD

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1. Project Name

ThreadSync

2. Project Owner

Shahariar Ratul. Employee PIN: 2347

3. Problem Statement

- In RMG manufacturing, **roughly 60–70%** of total lead time is spent not producing garments but tracking sample approvals (Proto → Fit → Size Set → PP → TOP) across buyer email threads, WhatsApp/Teams messages, and personal spreadsheets.
- Merchandisers manage many styles at once but have no central view: status lives in each person's Excel file, inbox, and notes. Follow-ups get missed, bottlenecks are hard to trace, and the same data is re-keyed repeatedly (interview finding: **500+ tech packs re-typed per season**).
- **The result is 15–24-day delays**, last-minute air-freight penalties, and management that cannot answer "where is style X right now?" without a phone call.
- Existing options are either informal and error-prone **(manual tracking)** or enterprise ERP/PLM systems that mid-tier factories and buying houses cannot afford or sustain.

4. Proposed Solution

A lightweight web app that gives merchandisers **one live board of the sample-approval pipeline**, kept up to date by AI reading the messages they already receive.

- **Pipeline board:** Tech Pack Received → In Development → Submitted to Buyer → Awaiting Buyer Comments → Revision Required → Approved → Bulk Ready. One card per style × sample type.
- **Three inputs, no behaviour change:** (1) forward buyer emails to a workspace address, (2) paste WhatsApp/Teams snippets, (3) type meeting or briefing notes. Buyer PLM/portal integration is a later phase.

- **AI does the tedious part:** identifies which style the message is about, extracts the decision, requested changes, dates, and next action, and moves the card — with the exact sentence it relied on shown as evidence.

- **Human stays in control:** anything the AI is unsure about, any backward move (e.g., Approved → Revision Required), and any date conflict goes to a *Needs Review* queue for one-click accept/edit/reject. Every change — AI or manual — is logged with its source.

- **Follow-ups never fall through:** each card carries a next action, owner, and due date; a rule engine flags overdue actions, samples awaiting buyer too long, and stale styles in a single *Exceptions* view.

The board answers, at a glance: current status of every style/sample · what is pending, delayed, or blocked · what action is next and who owns it · which follow-ups are due or overdue · where delays or inconsistencies are occurring · what information is missing.

5. Target Users

- **Primary:** merchandisers in mid-tier garment factories and buying houses (the people who own the buyer relationship and the spreadsheet today).

- **Secondary:** factory production planners (need to know when a sample is approved and what changed); merchandising managers (need a current, cross-buyer view without asking).

- **Indirect beneficiaries:** buyers, who get faster, more complete responses; BRAC IT's own enterprise clients in the RMG sector as a future product line.

6. Key Features

MVP Scop

- Workspace email forwarding address; paste-in messages; manual meeting notes
- Style & sample registry with CSV import from existing Excel trackers
- AI style matching and structured status extraction (stage, decision, changes, dates, next action, confidence, evidence quote)
- Confidence gating with a Needs Review queue; Unassigned inbox for unmatched messages
- Full audit log of every status change and its source
- Kanban board by sampling stage with drag-and-drop; style card with timeline, attachments, and open actions

- Filters and search (buyer, season, sample type, owner, flag)
- Exceptions view: overdue, awaiting buyer > N days, needs review, unassigned
- Next action + owner + due date on every card; hourly overdue / stale detection with in-app alerts

Should (if time allows): daily email digest · production hand-off notification · missing-info and date-inconsistency flags · manager summary view. Later phases: Gmail/Outlook auto-ingest · tech pack PDF extraction · AI-drafted buyer replies · buyer PLM/portal integration

7. AI Component

AI is used in exactly four places — all on text the merchandiser already has, all with human confirmation available:

1. **Style matching:** after deterministic style-code and thread matching fail, an LLM ranks registry candidates for the incoming message.

2. **Status extraction:** an LLM with an enforced JSON schema extracts sample type, stage, buyer decision, requested changes, dates, and a verbatim evidence quote from each email, pasted message, or note. Multiple styles in one email are handled as separate items.

3. **Next-action suggestion:** the same extraction proposes the follow-up task, likely owner, and due date, which the merchandiser confirms.

4. **(Should) Inconsistency and missing-info detection:** flags e.g. an approval with no PP sample recorded or conflicting dates across messages.

Guardrails: confidence threshold (0.8) for auto-apply; evidence quote must exist verbatim in the source; backward moves and date conflicts always require review; the AI never sends email or overwrites a later manual change. A labelled test set of 40–60 emails is built on day 1 to measure accuracy throughout the sprint.

8. Expected Outcome / Impact

- Merchandisers spend minutes, not hours, per day updating trackers — status is captured from the email as it arrives.

- Zero silent misses: every pending buyer response and every promised action has a due date and surfaces automatically when overdue.

- "Where is style X?" is answered in seconds by anyone with access, including production and management, without a call or a spreadsheet hunt.
- Earlier detection of slow approvals reduces the delays that force air freight; even recovering a few days per late style is material at RMG margins.
- A traceable change log replaces "who said what, when" arguments between buyer, merchandiser, and factory.
- For BRAC IT: a concrete, demonstrable AI-native product for the country's largest export sector, validated with practitioners, that can grow toward PLM integration and predictive delay alerts.

9. Data / Resources Required

- **Data:** style/sample registry (from merchandisers' existing Excel trackers via CSV); forwarded buyer emails and attachments; pasted messages; meeting notes. Demo uses synthetic emails plus anonymised real samples from two garment-factory contacts.
- **AI:** an LLM API with structured/JSON output (Claude Sonnet-class or equivalent); ~2–4k tokens per email — negligible cost.
- **Infrastructure:** Supabase (Postgres, auth, file storage, cron); Vercel hosting; an inbound email service (Postmark or Mailgun) with one custom subdomain and DNS records; Resend/SES for outbound digests (Should).
- **Not required for MVP:** buyer PLM/portal credentials, factory ERP access, Gmail/Outlook OAuth approval.

10. Success Criteria

- **End-to-end demo:** a real forwarded buyer email moves the correct card to the correct stage, with evidence shown — no manual typing.
- **Extraction quality on the labelled test set:** ≥90% correct style match, ≥85% correct stage, <5% "confidently wrong" (auto-applied but incorrect). Anything below threshold lands in review, not on the board.
- **Trust:** every card move shows source and evidence; any AI move can be reverted in one click.
- **Practitioner validation if possible:** Two voluntary merchandiser contacts run a week of real styles through it and confirm it replaces their personal tracker for sampling.

Why Should You Build This?

Sample-approval tracking is the single biggest hidden delay in Bangladesh's largest export industry, and today it runs on personal Excel files and unread emails. ThreadSync is deliberately narrow — one workflow, three inputs merchandisers already use, AI doing the reading and humans keeping control. Traditional factories use their own ERP, but they rely heavily on manual tracking as the ERP systems are too complex & hard to maintain without any AI native value proposition. ThreadSync could be a better alternative to that.